

SOFTWARE REQUIREMENTS SPECIFICATION

Library Management System (LMS)

Version 1.0

2026

Field	Details
Document Title	Software Requirements Specification – Library Management System
Version	1.0
Status	Draft
Prepared By	Group 7
Date	2026

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Library Management System (LMS). The document serves as the authoritative reference for developers, testers, project managers, and stakeholders throughout the software development lifecycle.

1.2 Scope

The Library Management System is a web-based application designed to automate and streamline library operations. It covers:

- Cataloging and managing all library resources (books, journals, digital media)
- Member registration, management, and access control
- Book issuing, returning, and renewal workflows
- Fine calculation and payment tracking
- Search and discovery of library resources
- Reporting and analytics for library administrators

1.3 Definitions & Acronyms

Term	Definition
LMS	Library Management System
Member	Registered library user with borrowing privileges
Librarian	Staff member managing day-to-day operations including circulation and resource cataloging
Administrator	System admin with full access and configuration rights
ISBN	International Standard Book Number
OPAC	Online Public Access Catalog — public-facing search interface
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface

1.4 Intended Audience

- Software developers and architects building the system
- QA engineers designing test plans
- Project managers overseeing delivery
- Library staff (Librarians and Administrators) who will use the system
- Stakeholders reviewing project scope and approval

2. Overall Description

2.1 Product Perspective

The LMS is a standalone web application that can optionally integrate with external systems such as payment gateways, barcode/RFID scanners, and email/SMS notification services. It replaces manual, paper-based library processes with a centralized digital platform accessible via any modern browser.

2.2 User Classes & Characteristics

User Type	Description	Technical Level
Member / Patron	Students, faculty, or public users who search and borrow books	Low
Librarian	Staff who issue, receive, return, and catalog all library resources	Medium

User Type	Description	Technical Level
Administrator	IT/library managers who configure the system and manage users	High

2.3 Assumptions & Dependencies

- The system will be deployed on a web server accessible via HTTPS.
- Users will access the system using modern browsers (Chrome, Firefox, Safari, Edge — latest 2 versions).
- A stable internet connection is required for web-based access.
- The library will provide barcode scanners or RFID readers for physical book check-in/out (optional hardware).
- Email/SMS gateway credentials will be provided for notification services.
- Initial data migration from existing records (if any) is a separate workstream.

2.4 Constraints

- The system must comply with applicable data privacy regulations (e.g., GDPR, local data protection laws).
- The system must support a minimum of 10,000 book records and 5,000 active members.
- Response time for search queries must not exceed 2 seconds under normal load.
- All financial transactions must be logged and auditable.

3. Functional Requirements

3.1 User Authentication & Authorization

ID	Requirement	Priority
FR-AUTH-01	The system shall support role-based login for Members, Librarians and Administrators.	High
FR-AUTH-02	Users shall authenticate using email/username and password. Optional two-factor authentication (2FA) must be supported on sign up.	High
FR-AUTH-03	The system shall enforce session timeout after 15 minutes of inactivity.	Medium
FR-AUTH-04	Administrators shall be able to create, edit, deactivate, and delete user accounts.	High
FR-AUTH-05	Password reset via registered email or phone number shall be supported.	High
FR-AUTH-	The system shall log all login attempts (successful and failed)	Medium

ID	Requirement	Priority
06	with timestamps.	

3.2 Book & Resource Catalog Management

ID	Requirement	Priority
FR-CAT-01	Librarians shall be able to add new books/resources with details: title, author(s), ISBN, publisher, edition, year, category, language, and number of copies.	High
FR-CAT-02	The system shall support import of book data via CSV or via ISBN lookup (external API integration).	Medium
FR-CAT-03	Librarians shall be able to update, archive, and delete resource records.	High
FR-CAT-04	The system shall track the total copies, available copies, and copies currently issued for each resource.	High
FR-CAT-05	The system shall support multiple resource types: books, journals, magazines, DVDs, and e-resources.	Medium
FR-CAT-06	Each resource record shall have a unique barcode/accession number auto-generated.	High

3.3 Member Management

ID	Requirement	Priority
FR-MEM-01	Librarians shall be able to register new members with: name, ID/card number, contact info, address, member type, and photo.	High
FR-MEM-02	The system shall generate a unique membership ID and library card number for each member.	High
FR-MEM-03	Member accounts shall have an expiry date. The system shall notify members 30 days before expiry.	Medium
FR-MEM-04	Librarians shall be able to renew, suspend, or terminate membership.	High
FR-MEM-05	The system shall maintain a complete borrowing history per member.	High
FR-MEM-06	Members shall be able to view their own profile, borrowing history, and outstanding fines via the member portal.	Medium

3.4 Book Issuing & Returns

ID	Requirement	Priority
FR-ISS-01	Librarians shall issue books to members by scanning the barcode or searching by title/member ID.	High

ID	Requirement	Priority
FR-ISS-02	The system shall enforce borrowing limits based on member type (e.g., students: 2 books, faculty: 4 books).	High
FR-ISS-03	The system shall set and display the due date upon issuing a book.	High
FR-ISS-04	Members shall be able to request renewals via the member portal (subject to librarian approval).	Medium
FR-ISS-05	The system shall process book returns, update availability in real time, and calculate any applicable fines.	High
FR-ISS-06	The system shall prevent issuing books to members with overdue items or outstanding fines.	High
FR-ISS-07	The system shall support book reservations/holds for items currently checked out.	Medium

3.6 Search & OPAC

ID	Requirement	Priority
FR-SCH-01	All users (including unauthenticated visitors) shall be able to search the catalog by title, author, ISBN, subject, and keyword.	High
FR-SCH-02	Search results shall display availability status (Available / Checked Out / Reserved).	High
FR-SCH-03	Advanced filters shall include: category, language, publication year range, and resource type.	Medium
FR-SCH-04	The system shall support autocomplete suggestions in the search bar.	Low
FR-SCH-05	Members shall be able to place a hold on a book directly from search results.	Medium

3.8 Reporting & Analytics

ID	Requirement	Priority
FR-REP-01	Librarian shall access pre-built reports: Most Borrowed Books, Overdue Items, Active Members, Fine Collections.	High
FR-REP-02	Reports shall be exportable in PDF and Excel formats.	High

4. Non-Functional Requirements

4.1 Performance

- Search queries shall return results within 2 seconds for databases up to 50,000 records under normal load.
- The system shall support at least 200 concurrent users without performance degradation.
- Page load time shall not exceed 3 seconds on a standard broadband connection.
- Batch report generation shall complete within 30 seconds.

4.2 Security

- All data in transit shall be encrypted using TLS 1.2 or higher (HTTPS only).
- Passwords shall be stored using a strong hashing algorithm (bcrypt or Argon2).
- The system shall implement protection against SQL injection, XSS, CSRF, and other OWASP Top 10 vulnerabilities.
- Access to administrative functions shall be restricted by IP allowlist (optional, configurable).
- Sensitive data (member PII) shall be encrypted at rest.
- All administrative actions shall be logged in an immutable audit trail.

4.3 Reliability & Availability

- The system shall achieve a minimum uptime of 99.5% (excluding scheduled maintenance).
- Scheduled maintenance windows shall be communicated at least 24 hours in advance.
- Automated database backups shall run daily, with backups retained for 30 days.
- The system shall recover from a server crash and be back online within 15 minutes (RTO).

4.4 Usability

- The interface shall be intuitive enough for a librarian with basic computer literacy to complete common tasks within 2 hours of training.
- The system shall be fully responsive and usable on desktops, tablets, and mobile browsers.
- The system shall comply with WCAG 2.1 Level AA accessibility standards.
- All error messages shall be descriptive and guide the user toward resolution.

4.5 Scalability

- The system architecture shall support horizontal scaling to handle growth in users and data volume.
- The database shall support up to 500,000 resource records and 100,000 member records without structural changes.

4.6 Maintainability

- The codebase shall follow documented coding standards and include inline documentation.
- The system shall use a modular architecture to allow independent updates to modules (catalog, members, fines, etc.).
- A deployment pipeline (CI/CD) shall be set up for automated testing and releases.

5. System Interfaces

5.1 External Integrations

Interface	Type	Purpose	Priority
Email Gateway (SMTP / SendGrid)	API	Send automated notifications to members	High
SMS Gateway (Twilio / similar)	API	Send SMS alerts for overdue and holds	Medium
ISBN Lookup API (Google Books / Open Library)	REST API	Auto-fill book details on catalog entry	Medium
Payment Gateway (Mobile Account / Bank Account)	API	Process digital fine payments	Low
Barcode / RFID Scanner	Hardware	Quick scan for issuing and returning books	Medium

5.2 Technology Stack (Recommended)

Layer	Recommended Technology
Frontend	React.js (responsive web UI)
Backend	Node.js (Express) or Python (Django/FastAPI)
Database	PostgreSQL (primary), Redis (caching & session management)
Authentication	JWT tokens + OAuth 2.0
Hosting	Cloud (AWS) or on-premises server
Search	Elasticsearch or PostgreSQL Full-Text Search

Note: The specific technology stack is subject to finalization based on team expertise and organizational standards.

6. Key Use Cases Summary

Use Case ID	Name	Actor	Brief Description
UC-01	Member Login	Member / Librarian /	User logs in with credentials; system

Use Case ID	Name	Actor	Brief Description
		Admin	validates and grants role-based access.
UC-02	Search Catalog	Any User	User enters search terms; system displays matching results with availability.
UC-03	Issue Book	Librarian	Librarian scans book and member card; system issues book and sets due date.
UC-04	Return Book	Librarian	Librarian scans returned book; system updates availability and calculates fine if overdue.
UC-05	Reserve Book	Member	Member places a hold on a checked-out book; system notifies when available.
UC-06	Pay Fine	Member / Librarian	Member or librarian records fine payment; system marks fine as settled.
UC-07	Add Book to Catalog	Librarian	Librarian enters or imports book details; system creates and stores catalog record.
UC-08	Generate Report	Administrator	Admin selects report type and date range; system generates and exports report.

7. Acceptance Criteria

The system shall be considered ready for release when all of the following are met:

- All High-priority functional requirements pass their corresponding test cases.
 - System handles 200 concurrent users with search response time under 2 seconds (verified by load testing).
 - Security audit confirms no critical or high-severity OWASP Top 10 vulnerabilities.
 - All automated notifications are delivered within 5 minutes of the triggering event.
 - User Acceptance Testing (UAT) is completed by at least 3 librarians with a satisfaction score of 4/5 or higher.
 - All reports export correctly in PDF and Excel formats.
 - Data backup and restore procedure is tested and documented.
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8. Appendix

8.1 Requirement Priority Legend

Priority	Definition
High	Core functionality — must be included in the initial release. System is unusable without it.
Medium	Important — should be included in the initial release. Workarounds exist but are not acceptable long-term.
Low	Nice to have — can be deferred to a future release without impacting core operations.

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